

KARTIKEYA VEMULA

Analytics Engineer | dbt | BigQuery | GCP Certified | Power BI (PL-300)
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Open to relocate across India

PROFILE

Analytics engineer building trusted data systems — dimensional warehouses, tested dbt layers, and decision-ready BI, now extended with streaming ingestion and LLM-assisted cleaning at the point of entry. GCP Professional Data Engineer and PL-300 certified, M.S. Data Science.

CORE SKILLS

Analytics Engineering: dbt Core, Dimensional Modeling, ELT/ETL & Streaming Pipelines, Data Quality Testing, Semantic Layer
Cloud & Warehousing: Google Cloud Platform, BigQuery, Snowflake, PostgreSQL, Redpanda (Kafka API), Docker, GitHub Actions
SQL & Python: SQL, Python, scikit-learn, XGBoost, Anomaly Detection, Root Cause Analysis
BI & Visualization: Power BI, Looker Studio, Streamlit, Plotly Dash
Engineering Practices: Git, CI/CD, Data Governance, LLM-Assisted Data Cleaning, A/B Testing

EXPERIENCE

Graduate Research Assistant, Risk Analytics — UMBC, Baltimore

Oct 2025 – Jan 2026

Project: Extreme Value AQI Risk Modeling

- Built an EPA AirData ETL pipeline producing a validated 6,573-record dataset (2006–2023) as the single trusted source for downstream risk modeling.
- Modeled pollutant risk with XGBoost (RMSE 8.99, R^2 0.852), establishing a defensible baseline for expected AQI behavior.
- Quantified a 6.65% conditional tail-risk probability across 98 exceedance events using Extreme Value Theory, delivered as reporting a non-technical monitoring team could act on directly.

Machine Learning Research Assistant — NIT Puducherry, Karaikal

May 2023 – Jul 2023

Project: Fish Detection & Recognition System

- Engineered an end-to-end computer vision pipeline in TensorFlow/Keras classifying four fish species as a single reproducible workflow.
- Removed class imbalance by standardizing a 572-image dataset to 143 images per class through augmentation, so accuracy became a meaningful metric.
- Improved generalization on a small dataset using transfer learning with pre-trained CNN architectures, controlling overfitting.

PROJECTS

Customer Intelligence Data Warehouse

Python, dbt, PostgreSQL, Redpanda, Gemini, Docker, Power BI

- Converted noisy retail data into a governed warehouse — 20 dbt models, 57 passing tests — and a streaming AI-clean gate that abstains on ambiguity, lifting valid-event acceptance from 55.6% to 85.2% with no records silently dropped.
- Modeled cohort retention and cumulative LTV by first-purchase month, converting a flat 67.25% repeat-rate into a time-decay view that pinpoints when each cohort's value plateaus and where retention spend still moves the needle.

Instacart Intelligence Platform

Python, dbt, PostgreSQL, BigQuery, Docker, Power BI

- Engineered a dimensional warehouse over 3.4M+ orders and 33.8M order-item rows, hardened into 19 dbt models with 238 automated tests passing at zero errors.
- Segmented 206,209 customers into four behavioral cohorts and quantified retention falling to 53.70% by order 10, converting a vague churn concern into a targetable at-risk population.

KPI Reliability & Diagnostic Engine

Python, dbt, SQL, Streamlit, GitHub Actions

- Benchmarked three anomaly-detection approaches against labeled ground truth and cut false positives from 2,958 to 563 (–81%) while recall held at 100%.
- Built contribution-share root cause analysis that isolated one region-device-browser segment as 100% of a 44.6% conversion-rate shortfall, plus a 9-check data quality gate blocking inference on failure.

CERTIFICATIONS

Google Cloud Certified: Professional Data Engineer • Google Cloud Certified: Associate Cloud Engineer • Microsoft PL-300: Power BI Data Analyst Associate • dbt Fundamentals • Snowflake Data Warehousing Workshop • Cisco Data Analytics Essentials • Kaggle Advanced SQL • Google Analytics Certification

EDUCATION

M.S. Data Science — University of Maryland, Baltimore County (UMBC), USA

2024 – 2026

GPA 3.78 / 4.0 | Focus: Machine Learning, Statistical Modeling, Experimental Design, Time Series Analysis

B.Tech Computer Science (AI & ML) — GITAM University, India

2020 – 2024

8.89 GPA, First Class with Distinction

EXTRA-CURRICULAR & LEADERSHIP

- Mentored peers in Mathematics, Digital Logic, Data Communication, and Data Analysis & Machine Learning.
- Led a 3-part UMBC peer-teaching series on ML terminology, model training, and real-world case studies.